

LUCA ZULIAN

Senior Backend Engineer

+39 347 529 4448
Milan, Italy
info@lucazulian.it
linkedin.com/in/zulianluca
github.com/lucazulian

PROFESSIONAL SUMMARY

Senior Backend Engineer with 20+ years of experience designing and delivering scalable, high-availability distributed systems and backend services. Deep expertise in functional programming with hands-on experience in Elixir, Rust, Python, Haskell, and more. Specialised in microservices architecture, event-driven systems (CQRS/Event Sourcing), real-time data pipelines, and cloud-native infrastructure on AWS. Experienced in integrating AI and large language models (LLMs) into production applications. Strong advocate for open-source software, CI/CD, observability, and engineering best practices including continuous delivery, defensive programming, and automated testing. Passionate about compilers, interpreters, domain-specific languages (DSLs), virtual machines, and autonomous systems.

TECHNICAL SKILLS

Programming Languages: Elixir, Rust, Python, Haskell, TypeScript, Java, C, C#, OCaml, Erlang, Scheme

Frameworks & Libraries: Phoenix, Ecto, Commanded, ASP.NET, Node.js, Xamarin

Cloud & Infrastructure: AWS, Terraform, Docker

Databases & Messaging: PostgreSQL, MongoDB, RabbitMQ

Methodologies: Microservices, CQRS, Event Sourcing, Domain-Driven Design (DDD), Test-Driven Development (TDD), CI/CD, A/B Testing, Observability

AI & Machine Learning: LLM Integration, Anthropic Claude, LangChain, OpenClaw, OpenCode, AI-powered feature development

EXPERIENCE

06/2024
Present
full remote

Senior Backend Engineer

Sibill (FinTech)

Contributed to automating financial operations for thousands of Italian SMBs, simplifying cash flow management, e-invoicing, and payment scheduling. Developed and optimised backend infrastructure in Elixir, ensuring robustness, scalability, and reliability of Sibill's core platform. Leveraged AI and integrated large language models (LLMs) directly into the product to automate workflows and deliver smarter, more intuitive user experiences.

Elixir / TypeScript / AWS / LLM / AI

02/2024
05/2024
full remote — short-term
engagement

Senior Backend Engineer

Remote (HR Tech)

Supported Remote.com's mission to simplify global employment by enabling companies to hire and manage talent across borders. Developed and enhanced backend infrastructure for the expenses product in Elixir, building scalable, reliable, and compliant systems. Streamlined employee expense management workflows for companies operating at a global scale.

Elixir / AWS

04/2022
01/2024
full remote

Senior Software Developer

Forza Football (Sports Tech)

Developed and maintained a high-throughput real-time data ingestion system in Elixir, processing live football match updates from multiple external data providers at scale. Contributed to and maintained several open-source Elixir libraries developed by the company.

Elixir / AWS / Terraform

04/2020
03/2022
Milan — hybrid remote

Software Developer

Casavo (PropTech)

Built backend infrastructure and microservices from scratch in Elixir and Rust for a real estate listing platform and an internal transaction management tool. Implemented the core domain using the Commanded library following CQRS and Event Sourcing patterns. Established and evolved Test Automation, CI/CD pipelines, and Observability practices.

Elixir / Rust / TypeScript / Python

06/2018
03/2020
Milan — full time

Software Developer

Prima Assicurazioni S.p.A. (InsurTech)

Developed new features and maintained an insurance back-office platform across a polyglot stack including Elixir, PHP, Elm, and Haskell. Improved system reliability and code quality through active refactoring and adherence to engineering best practices.

Elixir / PHP / Elm / Haskell

09/2016 05/2018 Milan — full time	Software Developer Implemented new features and maintained a performance-critical financial back-office platform in Python and Java. Designed and built an intraday financial market data collection system, aggregating feeds from multiple external data providers. Python / Java / Bash / C	Deus Technology (FinTech)
01/2016 08/2016 Milan — contract	Software Developer Analyzed, designed, and developed a digital rewards and gifting platform using Microsoft technologies (C#, .NET). Provided on-site technical consulting at ABB Automation. C# / .NET / JavaScript	ECRM Italia (MarTech)
04/2015 01/2016 Milan/Bologna — full time	.NET Developer / Consultant Designed and developed RESTful APIs for the internal orders team using ASP.NET Web API, a messaging system with RabbitMQ, and MongoDB. Led the migration to a new IBM inventory and e-commerce system in Java and Scala. Served as tech lead for the lost-and-returns, payment, and fraud detection teams. C# / .NET / Java / Scala / JavaScript	YOOX Group (Fashion E-commerce)
03/2012 03/2015 Milan — full time	Software Developer / Architect Developed web applications using ASP.NET and various JavaScript frameworks. Built cross-platform mobile applications for iOS, Android, and Windows Phone using Xamarin and native languages (Objective-C, Java, C#). Designed and developed a web-based application using Node.js and Meteor.js. Speaker at industry events and meetups. C# / .NET / Xamarin / Node.js / JavaScript	Dev4Side (Software Consultancy)
05/2011 02/2012 Padova — full time	Junior Software Developer Designed and developed websites and web applications in C# using ASP.NET (WebForms and MVC), deployed on both Windows and Linux (Mono) environments. C# / .NET / JavaScript / Mono	Sia-Net S.r.L (Web Agency)
11/2008 04/2011 Treviso — full time	Junior Software Developer Analyzed business requirements and developed customized ERP software applications covering sales, production, purchasing, planning, inventory management, and management control. Built small web and mobile applications, managed virtual infrastructure (VMware), and provided Help Desk and technical support. VB6 / VB.NET / .NET / C	Ergon Informatica (IT Consultancy)
04/2004 10/2008 Padova — full time	PLC Manager — Industrial machine division Automated industrial processes and programmed anthropomorphic robots (ABB, FANUC, KUKA) for product movement and handling. Applied FANUC TP/KAREL, ABB RAPID, KUKA KRL, and Siemens PLC languages (KOP, FUP, AWL). Conducted R&D for new industrial automation solutions. C / VB.NET / Assembly	Digipack S.r.L (Industrial Automation)

EDUCATION

1999 2004 Treviso	Electrical and Advanced Industrial Automation — 100/100	Istituto Tecnico Industriale "E.Barsanti"
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CERTIFICATIONS

2021 Online	Decision-making for Software Development Teams Workshop ID: 2021 07 01	Avanscoperta
2021 Online	Applied Functional Programming Workshop ID: 2021 06 23	Avanscoperta
2021 Online	EventStorming Remote Modelling Workshop ID: 2021 05 11	Avanscoperta
2021 Online	Haskell Fundamentals	47 Degrees
2022 Online	Microservices Practical Workshop ID: 2022 01 21	Avanscoperta

OTHER ACTIVITIES

2019
Present
Worldwide

Autonomous & Self-Driving Robots

Personal Project

Designing and building autonomous and self-driving robots as a personal passion project. Lego Robot Champion — competed and won in international Lego robotics competitions.

Robotics / Autonomous Systems / Embedded C

PERSONAL PROJECTS

04/2026
Present
Open Source

Founder & Developer

BoringlyRSS (SaaS / Dev Tool)

Designed and built a full-featured RSS/Atom feed reader from scratch, documented publicly via a build-in-public blog. Features a reliable feed-fetching pipeline (Oban workers, RabbitMQ, ETag/Last-Modified conditional fetching), full-text search with PostgreSQL `pg_trgm`, AI summarisation via local LLM inference (Ollama), per-user MCP endpoints for LLM clients (Claude, Cursor, Zed), OAuth 2.0/OpenID Connect server (`boruta_auth`), WebAuthn/passkey authentication (`wax_`), newsletter inboxes, podcast support, reading statistics, and a Progressive Web App.

Elixir / Phoenix / LiveView / PostgreSQL / RabbitMQ / Oban / Ollama / MCP / OAuth 2.0 / WebAuthn

LANGUAGES

Italian — native
English — proficient

HOBBIES

Tech speaking, blogging, running, and participating in various meetups and user groups.

OPEN SOURCE

Actively contributing to open-source projects on GitHub.